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Sequence and Phylogenetic Analysis of the Untranslated Promoter Regions for *HLA* Class I Genes

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Abstract

Polymorphisms located within the *MHC* have been linked to many disease outcomes by mechanisms not yet fully understood in most cases. Variants located within untranslated regions of *HLA* genes are involved in allele specific expression and may therefore underlie some of these disease associations. We determined sequences extending nearly 2kb upstream of the transcription start site for 68 alleles from 57 major lineages of classical *HLA* class I genes. The nucleotide diversity within this promoter segment roughly follows that seen within the coding regions, with *HLA-B* showing the highest (~1.9%), followed by *HLA-A* (~1.8%), and *HLA-C* showing the lowest diversity (~0.9%). In spite of its greater diversity, *HLA-B* mRNA expression levels determined in 178 European Americans do not vary in an allele- or lineage-specific manner, unlike the differential expression levels of *HLA-A* or *HLA-C* reported previously. Close proximity of promoter sequences in phylogenetic trees is roughly reflected by similarity of expression pattern for most *HLA-A* and *-C* loci. While promoter sequence divergence might impact promoter activity, we observed no clear link between the phylogenetic structure as represented by pairwise nucleotide differences in the promoter regions with estimated differences in mRNA expression

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Conflict of Interest

The authors declare no conflict of interest.

levels for the classical class I loci. Further, no pair of class I loci showed coordinated expression levels, suggesting that distinct mechanisms across loci determine their expression level under non-stimulated conditions. These data serve as a foundation for more in depth analysis of the functional consequences of promoter region variation within the classical *HLA* class I loci.

Keywords

MHC; HLA-Class I; gene regulation; expression variation; phylogenetics; 5'UTR

Introduction

The *MHC* is the most polymorphic locus in the human genome (1, 2), and it is also the richest segment genome-wide in terms of disease associations, including bacterial, viral, inflammatory and autoimmune diseases, cancer, neurological disorders, and drug hypersensitivity (3). Assigning the mechanism for these disease associations is confounded by the extensive polymorphism, dense gene content with strong linkage disequilibrium between pairs of genes, and functional similarity of the molecules encoded by genes within the *MHC*.

Several studies have indicated that expression levels of certain *HLA* loci vary in an allotype dependent manner (4–16) and this variation has been shown to associate with certain diseases, such as outcome after HIV infection (4, 17), Crohn's disease (4), HBV clearance (8), graft-versus-host disease after unrelated hematopoietic cell transplantation (18, 19), Parkinson's disease (20), systemic lupus erythematosus (21) and certain cancers (22–24). The regulation of *HLA* class I expression is mediated by a series of core promoter motifs, which are located within 500 bp upstream of the start site and are relatively conserved (25). Binding sites for regulatory proteins further upstream from the core promoter have also been identified (26, 27) and it is likely that others exist, some of which may be polymorphic. Indeed, genetic variants and epigenetic modifications located within the 3' and 5' untranslated regions (UTR) (6, 13, 25, 28, 29), as well as the coding region (14), have been shown to alter *HLA* allele expression levels, but so far, these mechanisms account for only a small percentage of the expression variation found within the respective genes.

Here we provide extensive allelic sequence information for 5' UTR and upstream promoter regions of the classical *HLA* class I genes and the homologous region for one non-classical gene, *HLA-E*. Based on these sequences, the intra-locus nucleotide diversity and the phylogenetic congruence between the 5' UTR promoter and the gene region sequences for each of the three classical class I loci were determined. Expression variation observed among the 57 *HLA* class I lineages was examined in relation to promoter sequences. These data serve as a basis for functional studies aimed at identifying the variants that are involved in differential *HLA* allelic expression levels.

Materials and Methods

Sequencing

Cell lines, homozygous at the *HLA* class I loci, from the Tenth International Histocompatibility Workshop were selected for *HLA* class I promoter sequencing (30). These cell lines represent common alleles that predominantly originate from European American (EA) ancestry. We also selected common African alleles for promoter sequencing from homozygous donors recruited from the Frederick National Laboratory for Cancer Research Donor Program. At least two donors homozygous for a particular allele were selected for study.

Primers specific for each *HLA* gene interrogated (*HLA-A*, *-B*, *-C* and *-E*) were designed using online Primer 3 software (31). The targeted region covering approximately 2 KB upstream of the transcription start site was amplified with locus specific primers (Supplementary Table 1) using the following PCR conditions; 95°C for 5 min, 40 cycles of 95°C for 15 s, 60/63°C for 15 s and 72 °C for 1 min, finally 72 °C for 1 min. An annealing temperature of either 60 or 63 °C was used depending on the locus. The purified PCR products were sequenced in both directions with at least 2 overlapping primer sets using the ABI-3130XL DNA analyzer (Applied Biosystems).

Sequences were analyzed using Sequencher® (Gene Codes) and Mutation Surveyor® (Soft Genetics) software and aligned with BioEdit (<http://www.mbio.ncsu.edu/bioedit/bioedit.html>) and Clustal. All donors homozygous for a specific *HLA* coding region were also homozygous in the 5'UTR sequence. Previously identified transcription factor binding sites were highlighted using the BioEdit program.

Putative transcription factor binding sites were obtained using the online software Alibaba 2.0 (32) with default settings.

Sequences were deposited to Genbank database with the following accession numbers KY427751-KY427809 (<https://www.ncbi.nlm.nih.gov/genbank/>).

HLA-B and *-E* mRNA expression levels using Real Time PCR

Healthy EA donors from the Research Donor Program at the Frederick National Laboratory for Cancer Research were utilized to measure the *HLA-B* (n=178) and *HLA-E* (n=27) mRNA expression levels using quantitative PCR (qPCR). Primers are located in exon 4 and exons 2–3 for *HLA-B* and *-E*, respectively. *HLA-B*, *-E* and the housekeeping gene *beta-2 microglobulin* (*β2M*) were amplified with the PCR conditions previously described (5, 6). Primer specificity was verified by both melt curve analysis and sequencing of *HLA-B* and *-E* PCR amplicons. PCR amplification efficiencies were tested for all common *HLA-B* (n=18) and *-E* (n=2) allelic lineages. *HLA-B* and *-E* expression levels were normalized to *β2M* and calculated using the $2^{-\Delta\Delta C_t}$ method. Normalizing *HLA-B* and *-E* expression levels by the housekeeping genes *GAPDH*, *18s* and *β-Actin* show very consistent results to those using *β2M* (data not shown).

Primer-BLAST and sequencing of PCR amplicons were performed to verify that only the specific targeted lineage corresponding to each given pair of *HLA-B* and *-E* primers was amplified. To further confirm accuracy of the *HLA-B* primers, an additional set of primers within the exon 5–7 region of the *HLA-B* locus was designed. The PCR amplification efficiencies and *HLA-B* specificity for the exon 5–7 primer set were shown to be equal across *HLA-B* allelic lineages.

Phylogenetic analysis

A comprehensive alignment including the sequences generated in this study, full length genomic sequences obtained from the IMGT/HLA database (1), and sequences identified by BLAST searches of nucleotide databases was constructed for phylogenetic analyses. The completed alignment consisted of up to 5.3kb, which encompassed promoter (~2kb) and coding regions (~3.3kb) for 68 alleles from 57 lineages of the three classical *HLA* class I loci. Phylogenetic trees were constructed using the neighbor-joining method (33) based on nucleotide distances calculated by Kimura's 2-parameter method (34) across relevant segments of the alignment. Gaps were excluded on a pairwise basis.

Nucleotide diversity for homologous positions within the promoters of *HLA-A*, *-B* and *-C* was estimated using Kimura's 2-parameter method (34) and examined by looking at intra-locus nucleotide diversity within small (150bp) windows with sliding window analysis across the alignment as previously described (35). Single representatives from the 57 distinct lineages of *HLA* class I genes were included in the sliding window analysis.

Statistical analyses

The estimated mean expression levels of individual *HLA-B* (178 donors) and *HLA-C* (256 donors) lineages were calculated using the linear regression function (lm) of the R package (36) as previously described for *HLA-A* mRNA (6). All donors were of European descent. The magnitude and range of expression differences across the three loci are likely related to consequent phenotypic effects at each *HLA* locus. Although standard units are used to measure intralocus expression differences, comparisons between loci require a normalization procedure to correct for the different experimental measurement procedures developed for each locus. To standardize the measurements across loci, the coefficient of variation (CV) was determined by estimating the ratio of the standard deviation to the mean of expression in a cohort of individuals typed at each of *HLA-A*, *-B* and *-C* (n=172). The CV can be either directly estimated from each of the 172 individual measurements of expression at each locus or alternatively, they can be estimated by utilizing mean effects of allelic lineages. In the first approach, expression was experimentally determined by qPCR for every individual at each locus allowing direct calculation of means, variances and CV. For the second approach, expression was inferred by combining the mean effects of allelic lineages that make up the known genotypes of each individual. The individual's expression level is thus determined from the standard expression levels of the alleles making up his/her genotype. The latter approach should be less sensitive to the increases in CV due to experimental errors.

Spearman correlations and Mann-Whitney tests were performed using GraphPad Prism software version 6 (GraphPad Software, San Diego California USA).

Results

A region of 1.8–2kb extending upstream of the translation start site (TSS) was sequenced for a total of 68 alleles from the three *HLA* classical class I genes; 19 *HLA-A*, 33 *HLA-B*, and 16 *HLA-C* alleles. The 68 alleles group into 57 distinct allelic lineages, comprising 17 of 21 known lineages at *HLA-A*, 27 of 36 known lineages at *HLA-B*, and 13 of 16 known lineages at *HLA-C*. Two or more donors homozygous for each allele were sequenced and covered the common allotypes present within individuals of European and African descent. A previous report provided sequences 859bp upstream of the TSS for the 16 *HLA-C* lineages (37), which overlaps with about half of that provided herein, and these 859bp sequences are identical to those determined in this study for the same segment of DNA.

Variation in transcription factor binding sites (TFBS) of the *HLA* gene core promoter (38–42) was identified between allelic lineages of *HLA-C* (Fig. 1A), where *HLA-C*03*, *-C*07* and *-C*17*, which were previously identified as having low expression levels (4), possess variation in at least one of the eight *HLA-C* core promoter motifs. Relative to consensus, the *HLA-C*03* lineage contained a single nucleotide change in the TATA binding site, while the *HLA-C*07* lineage contained a single nucleotide change within the Enhancer A (κ B2) binding site and a deletion polymorphism in the ISRE binding site. The *HLA-C*17* lineage contained single base changes within both the Enhancer A κ B1 and TATA binding sites, and two base changes within the κ B2 binding site. Furthermore, the *HLA-C*17* lineage possesses a 7bp deletion located at position –37 to –43bp upstream of the TSS, and an insertion of 6bp is observed at position –181 to –187 (Fig. 1A). The potential for these variants to explain the low cell surface expression of *HLA-C*03*, *-C*07* and *-C*17* will require experimental examination.

Among *HLA-A* lineages, the core promoter TFBS are highly conserved. A single variant located at position –208bp upstream of the TSS is observed in the Enhancer A (κ B1) binding site for the *HLA-A*30* lineage (Fig. 2A). Variation in the *HLA-B* core promoter TFBS is also limited with a few nucleotide changes in the ISRE, S (H)-box and X1/X2 binding sites (Fig. 3A). In general, however, the *HLA-B* promoter region as a whole is more polymorphic than that of *HLA-A* or *-C*, in keeping with the relative polymorphism across their coding regions.

The nucleotide diversity observed within the *MHC* region, ranging upwards of several percent for classical *MHC* genes, has previously been shown to be far higher than the genome average of ~0.1% (43). This holds true for the regulatory region ~2kb upstream of the initiation codon of the classical class I genes, as well, where the intra-locus nucleotide diversity across the region for the pool of *HLA-A* (Fig. 2B) *-B* (Fig. 3B) and *-C* (Fig. 1B) lineages is on average ~1.8%, ~1.9% and ~0.9% respectively, roughly following the diversity within the coding regions. Divergence across the three loci is also evident, in part due to the presence of insertion elements, which were identified in the UCSC browser (<https://genome.ucsc.edu>) as LINE elements MLT1E3 and L2 within *HLA-A*, and L1M4a1 within *HLA-B* and *-C*. Surprisingly, the most conserved regions for *HLA-B* and *-C* are not those of the core promoter region where multiple well-characterized TFBS are positioned, but rather, positions –700bp to –500bp are more conserved in *HLA-C* than any other 200bp

window examined across the remainder of the promoter region (generally above 1% diversity) (Fig. 1B and 3B). The lowest *HLA-B* diversity of about 0.2% is located from positions -1720bp to -1620bp (Fig. 3B) and that for *HLA-A* of about 0.5% is located near -500bp (Fig. 2B).

The phylogenetic tree of allelic lineages for the *HLA-A* gene constructed from promoter sequences is generally congruent with that found for the remainder of the gene region, indicating that recombination has not scrambled the sequences extensively. Seven promoter clades are identified (Fig. 4A), consisting of *A*02/A*34/A*68*, *A*25/A*26/A*66*, *A*23/A*24*, *A*31/A*33*, *A*29/A*32/A*74*, *A*01/A*11*; and *A*03/A*30* lineages. These clades are also recovered for the remainder of the gene (Fig. 4B), apart from the *A*34* lineage being located in the *A*25/A*26/A*66* clade. The relative position of the *A*23/A*24* clade also differs between promoter and gene coding segments. Linking previously published *HLA-A* mRNA expression levels (6) with the phylogenetic trees indicated that certain lineages with similar or identical promoter regions possess similar mRNA expression levels, such as *A*23/A*24* and *A*31/A*33* (Fig. 4C), whereas others, such as *A*03* and *A*30*, do not. *A*03* and *A*30* show a 10-fold difference in average lineage expression levels, which may result in part from the eight variable sites observed between the two lineages within the ~2Kbp promoter region (Supplementary Table 2).

Nine clades were identified within the *HLA-B* promoter tree (Fig. 5A): *B*35/B*51/B*52/B*53/B*58*, *B*18/B*27/B*37*, *B*40/B*41/B*45/B*49/B*50*, *B*15/B*46*, *B*55/B*56*, *B*13/B*44/B*47*, *B*14/B*38/B*39*, *B*07/B*08/B*42*; and *B*57* (the latter being the only lineage-specific clade). The tree derived from coding sequences recovers the *B*35/B*51/B*52/B*53/B*58* clade with incorporation of the *B*57* lineage into the clade (Fig. 5B). The *B*15/B*46* and *B*55/B*56* pairings also reflect the promoter clades. However, the promoter-based *B*40/B*41/B*45/B*49/B*50* clade splits, where *B*40/B*41* groups with *B*07/B*08/B*42* in the coding region. Likewise, where the promoter forms a *B*13/B*44/B*47* clade, *B*27/B*44/B*47* form a clade in coding regions and *B*13* forms an isolated tree branch. These and other inconsistencies between the promoter and coding region trees can be explained by infrequent recombination events that have displaced some, but not all relationships. A parsimonious reconciliation of the trees requiring the fewest recombination events suggests that lineages sharing inconsistency arose after their common ancestral sequence underwent recombination, giving rise to novel associations across promoter and coding regions.

HLA-B mRNA expression levels were determined in 178 European American (EA) donors, and unlike allele-specific differential expression at *HLA-A* and *-C*, we observed very minor differential expression across alleles (Fig. 5C). A fold change of only 1.2 was observed between the coefficient of variation expression levels of the highest (*B*13*) and the lowest (*B*52*) lineages at *HLA-B* (Fig. 5C and 5D), whereas the fold change within the *HLA-A* and *-C* loci is 3.8 and 2.2 fold, respectively, when comparing the highest and lowest coefficient of variation for expression levels at each of these loci.

HLA-B mRNA expression levels were measured using a set of qPCR primers that are specific only to *HLA-B* lineages, as determined by sequencing and online primer BLAST

software (data not shown). The primer set, which targets exon 4, was shown to amplify all common *HLA-B* lineages with equal efficiency (Supplementary Figure 1A), and no correlation was observed between PCR efficiency of the primers for each *HLA-B* lineage and average expression level of each lineage (Spearman correlation coefficient; $R = 0.22$, $p = 0.36$) (Supplementary Figure 1A). A second set of qPCR primers spanning *HLA-B* exons 5–7 was generated, which amplified all *HLA-B* lineages with equal efficiency and validated the data obtained using exon 4 primers (data not shown). A significant positive correlation between *HLA-B* mRNA expression levels using exon 4 vs. exon 5–7 primer sets was observed for 61 EA healthy donors (Supplementary Figure 1B, $R = 0.81$, $p < 0.0001$). *HLA-B* mRNA expression levels were also shown to be quite stable in 30 EA donors for whom measurements were repeated three years after the initial measurement (Supplementary Figure 1C, $R = 0.72$, $p < 0.0001$).

The *HLA-C* phylogenetic trees for the promoter region and the coding region show very similar clade structure (Fig. 6A vs. Fig. 6B), the two main differences involving the *C*03* lineage, which groups with *C*01/C*04/C*14* in coding region, but forms a distinct (though weakly supported) clade in the promoter region, and *C*02*, which groups weakly with *C*15:02* in the coding region, but again forms a distinct weakly supported clade in the promoter region. Similar to *HLA-A* (6) and *HLA-B* (Supplementary Figure 1C), the mRNA expression levels of *HLA-C* are stable across time, as observed in 68 EA donors whose expression was measured three years apart (2012 and again in 2015) (Supplementary Figure 1D, $R = 0.62$, $p < 0.0001$). *C*01* and *C*14*, which are distinguished by only four nucleotides in the promoter region, are the two lineages with the greatest difference in mRNA expression levels (2 fold) within a promoter region clade (Supplementary Table 2). These two lineages differ at position –824, where an Sp1 is predicted for the *C*01* lineage (Alibaba 2.0 software (32)), but not the *C*14* lineage, which could explain the difference in their expression levels.

It is notable that despite the allele-specific variation in expression level for *HLA-A* and *HLA-C* mRNA (5, 6), the extent of promoter region variation between pairs of alleles overall does not correlate with differential mRNA expression levels of those alleles. There is no clear link between the phylogenetic structure as represented by pairwise nucleotide differences in the promoter region with estimated differences in mRNA expression levels observed for *HLA-A* (Fig. 7A, Spearman correlation coefficient; $R^2 = 0.003$), *HLA-B* (Fig. 7B, Spearman correlation coefficient; $R^2 = 0.0007$) and *HLA-C* (Fig. 7C, Spearman correlation coefficient; $R^2 = 0.025$). Note that the non-independence of pairwise comparisons in Figure 6 (due to shared phylogeny) indicates that the significance values are not valid. Nevertheless, these data suggest that only very few specific variants in the promoter region are likely to account for differential expression across lineages.

Promoter sequences for the *HLA-A*, *-B* and *-C* genes form distinct locus-specific clades (data not shown) in a manner that largely reflects locus-specificity of the coding region. The common ancestral locus that gave rise to *MHC-A*, *-B* and *-C* underwent duplication to give rise to two loci, one of which was the ancestral *MHC-A* while the other locus underwent duplication at a later date, giving rise to the ancestral *MHC-B* and *-C* loci (44). These relationships are reflected in the promoter tree where *HLA-A* branches outside of an *HLA-B* and *-C* clade and indicate that interlocus conversion events have not noticeably contributed

to the diversity of the promoter. This absence contrasts to findings in the 3'UTR where interlocus conversion between *HLA-B* and *HLA-C* is strongly implicated in miRNA induced control of expression (35).

Expression levels of *HLA-A*, *-B* and *-C* mRNA were measured in 172 EA individuals and tested for correlations between pairs of loci. Very poor correlations in expression levels were detected between the pairs (*HLA-A* and *-B*, Spearman correlation coefficient; $R^2 = 0.0002$, $p = 0.87$, Fig. 8A; *HLA-A* and *-C*, Spearman correlation coefficient; $R^2 = 0.02$, $p = 0.05$, Fig. 8B, and *HLA-B* and *-C*, Spearman correlation coefficient; $R^2 = 0.02$, $p = 0.07$, Fig. 8C). These data indicate that unlike the HLA-class II genes, which are regulated by common factors and display strong correlations in expression levels across loci (29, 45), under healthy conditions, the classical class I genes are regulated largely independently, presumably by at least some distinct mechanisms.

The coefficient of variation for expression of *HLA-A*, *-B* and *-C* was estimated in these 172 individuals for whom complete 3-locus data were available, and used to determine the relative variation in expression for the 3 loci. Both observed and expected CV, where expected values are based on regression estimates of allelic effects for each genotype, were determined. The CVs increased in order of *HLA-B* (lowest) to *HLA-A* (highest) and the ratio relative to *HLA-B* is 1 to 2.84 (*HLA-C*) to 3.5 (*HLA-A*) for observed, and 1 to 4 (*HLA-C*) to 5.6 (*HLA-A*) for expected expression. Observed expression fully incorporates experimentally-determined variation, whereas experimental “noise” can be minimized using expected variation.

No mRNA expression variation was observed between the only two common *HLA-E* alleles, *HLA-E*01:01* and *-E*01:03*, as measured by qPCR in 27 individuals (Supplementary Figure 1E, $p = 0.95$). The qPCR primers amplified both alleles with virtually the same PCR efficiency of 1.95 and 1.96, respectively. *HLA-E*01:01* and *-E*01:03* differ in the coding region by only a single nonsynonymous nucleotide change, and in keeping with their high sequence similarity, the ~1.2Kb sequence upstream of the TSS for these alleles was identical (data not shown).

Discussion

Differential *HLA* expression levels have recently been shown to associate with a number of diseases (4, 5, 8, 18, 19, 46), and the sequences provided herein serve as an initial step in identifying novel variants responsible for allele specific expression variation. In spite of substantial variation in the promoter regions of the *HLA* class I genes, which reflects that observed in the coding region of these loci, a Spearman correlation test revealed no strong correlation between similarity in promoter sequence and expression levels between pairs of alleles, suggesting that only a limited number of promoter region variants are likely to impact allele specific expression variation.

Three *HLA-C* lineages, *HLA-C*03*, *-C*07* and *-C*17*, possess the lowest expression levels and also contain variants within confirmed transcription factor binding sites of the core promoter. These three lineages all carry the intact miR-148a binding site in their 3'UTR,

which has been shown to reduce *HLA-C* expression levels (28). It is possible that the disrupted core promoter along with miR regulation of these three alleles result in their low expression levels relative to other *HLA-C* alleles, highlighting multiple distinct mechanisms determining differential expression of the *HLA-C* alleles.

Despite high levels of nucleotide diversity found in the *HLA-B* promoter region, we observe a low level of *HLA-B* expression variation across donors and find no evidence for allele-specific mRNA expression levels for *HLA-B*, further distinguishing *HLA-B* from *HLA-A* and *-C*. The stability of *HLA-B* mRNA expression levels measured three years apart indicates that expression differences across subjects are genuine and are not simply due to measurement fluctuation. The lack of allele-specific expression along with the stability of *HLA-B* mRNA expression levels suggests that trans, rather than *cis* polymorphism may modify *HLA-B* expression. Although we find no evidence for *HLA-B* lineage specific mRNA expression differences, recent reports suggest that cell surface expression levels of HLA-B do vary in a lineage-specific manner to some extent (47, 48), which may result from lineage-specific translational events. Using three antibodies (Tu149, B1.23.2 and 22E1), Chappell et al. showed that the surface expression levels of HLA-B*57, -B*27, -B*07 and -B*35 inversely correlated with promiscuity of peptide binding (48), where B*57 was expressed at the highest level, followed by B*27, B*07, and B*35 in descending order. A second study used the anti-Bw4 Ab clone 0007 to measure the surface expression of Bw4 alleles, including HLA-B*57 and -B*27 (47). The anti-Bw4 antibody, however, indicated that HLA-B*27 has higher expression than -B*57 (47). These studies were performed with different cell types, which may result in a distinct hierarchy of HLA-B allotype expression levels. As differential cell surface expression levels are likely to impact HLA-B restricted T cell responses, it will be important to pursue this line of study more fully.

Differential allele-specific expression levels have been observed at loci within the MHC, including the *TNF*, *LTA*, *HSP70*, *ZFP57* and *HLA-DRB1* genes (48–51). This regulation is thought to involve a complex interplay between genetic and epigenetic mechanisms (50), which is also likely to be true for *HLA-A*, where methylation levels are known to associate with differential expression levels of *HLA-A* alleles (6). Interestingly, methylation does not appear to regulate expression levels of *HLA-B* or *-C*.

A highly efficacious vaccine against SIV in rhesus monkeys has been shown to elicit MHC-E-restricted CD8⁺ T cell responses against a broad array of SIV peptides (49, 50). These responses are likely to explain in part the remarkable protection conferred by this vaccine, sparking interest in considering HLA-E in vaccine design as a mediator of protection against HIV-1. There are only two common alleles of *HLA-E*, *-E*01:01* and *-E*01:03*, which differ by a single amino acid (Gly to Arg at position 107, respectively). *HLA-E*01:03* is expressed at a higher level on the cell surface relative to *HLA-E*01:01* (15), which may affect their functional activity. We found no evidence for allele specific expression variation in *HLA-E* transcript levels and no difference in the sequence ~1.2kb upstream of the TSS between these two common lineages, suggesting that any variation in cell surface expression of the allotypes are due to post translational modification.

Regulation of *HLA* class I expression levels appears to occur at both the transcriptional and translational level, depending on the locus. The work described herein provides a foundation for greater understanding of the mechanisms responsible for differential regulation of class I allelic expression, which may lead to the ability to manipulate *HLA* expression levels in diseases where class I expression levels are known to associate with outcome.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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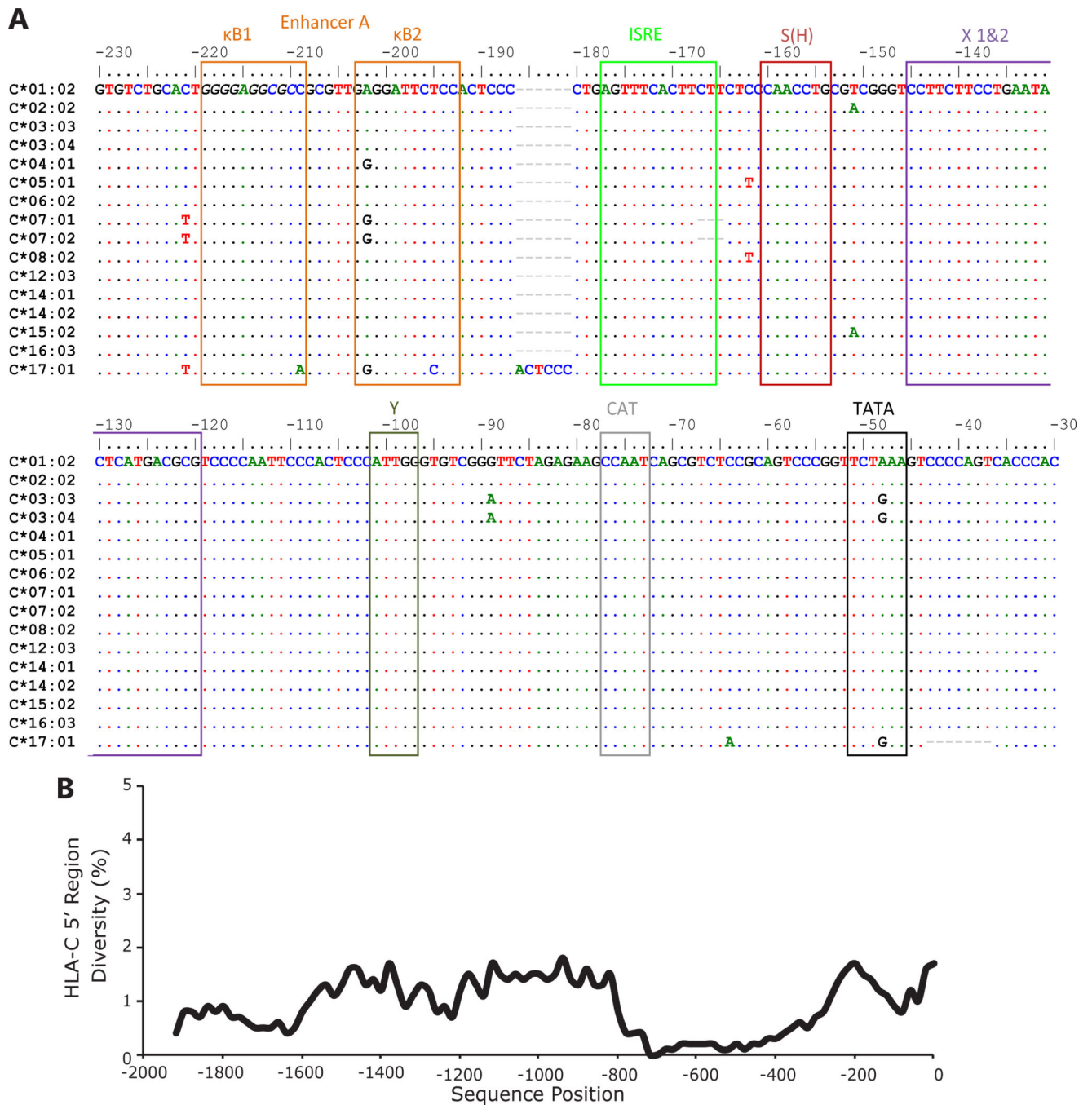
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**FIGURE 1.**

Transcription factor binding motifs in the *HLA-C* proximal promoter region and nucleotide diversity across ~2kb of the *HLA-C* promoter regions. (A) Highlighted regions represent confirmed transcription factor binding site motifs based on previously published data (38–42). Sequences for the *HLA-C* loci showing a fragment of the total 2kb upstream region sequenced. Dots represent consensus with the reference sequence, and dashes indicate a deletion. Sites are numbered according to distance from the translation start site (TSS). (B) Intra-locus *HLA-C* nucleotide diversity is examined within 150bp windows using a sliding

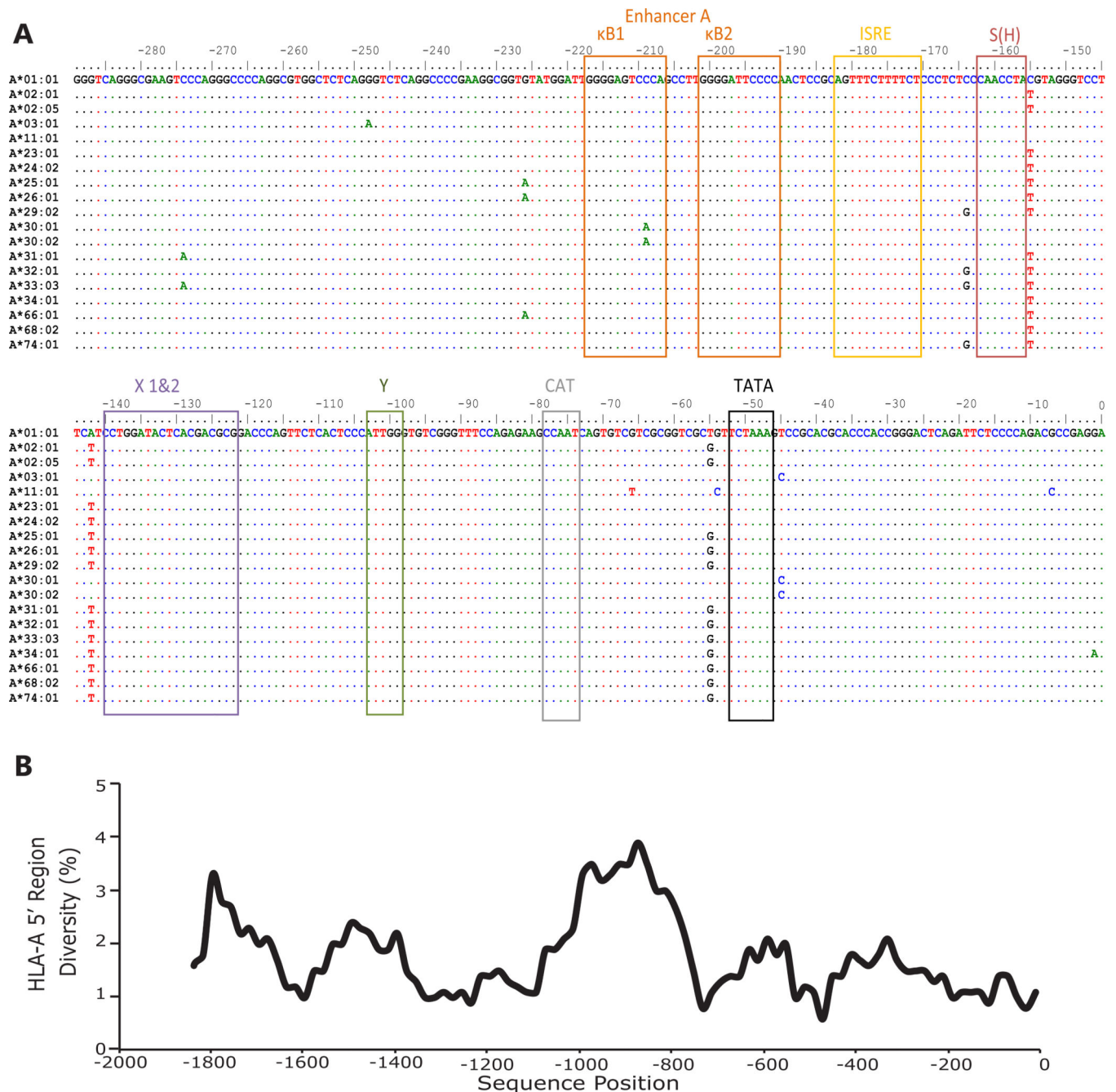
window analysis of aligned sequences consisting of a single representative of each allelic lineage (n=13). The x axis represents the nucleotide position in the alignment relative to the TSS (represented as 0).

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**FIGURE 2.**

Transcription factor binding motifs in the *HLA-A* proximal promoter region and nucleotide diversity across ~2kb of the *HLA-A* promoter regions. (A) Highlighted regions represent confirmed transcription factor binding site motifs based on previously published data (38–42). Sequences for the *HLA-A* loci showing a fragment of the total 2kb upstream region sequenced. Dots represent consensus with the reference sequence, and dashes indicate a deletion. Sites are numbered according to distance from the translation start site (TSS). (B) Intra-locus *HLA-A* nucleotide diversity is examined within 150bp windows using a sliding window analysis of aligned sequences consisting of a single representative of each allelic

lineage (n=17). The x axis represents the nucleotide position in the alignment relative to the TSS (represented as 0).

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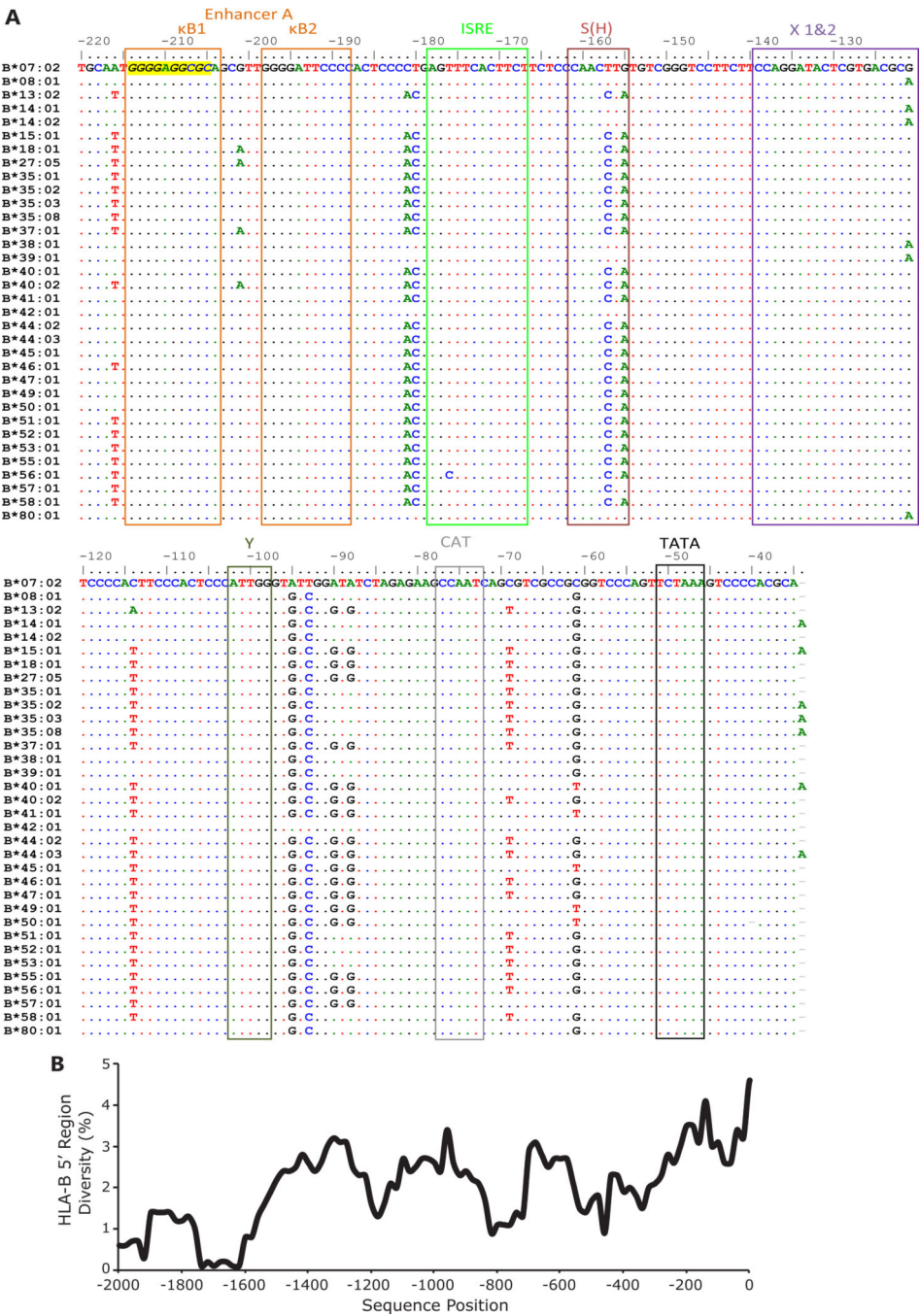


FIGURE 3. Transcription factor binding motifs in the *HLA-B* proximal promoter region and nucleotide diversity across ~2kb of the *HLA-B* promoter regions. (A) Highlighted regions represent confirmed transcription factor binding site motifs based on previously published data (38–42). Sequences shown are a fragment of the total 2kb upstream region sequenced. Dots represent consensus with the reference sequence, dashes indicate a deletion and yellow highlight represent sp1 binding site. Sites are numbered according to distance from the TSS. (B) Intra-locus *HLA-B* nucleotide diversity is examined within 150bp windows using a

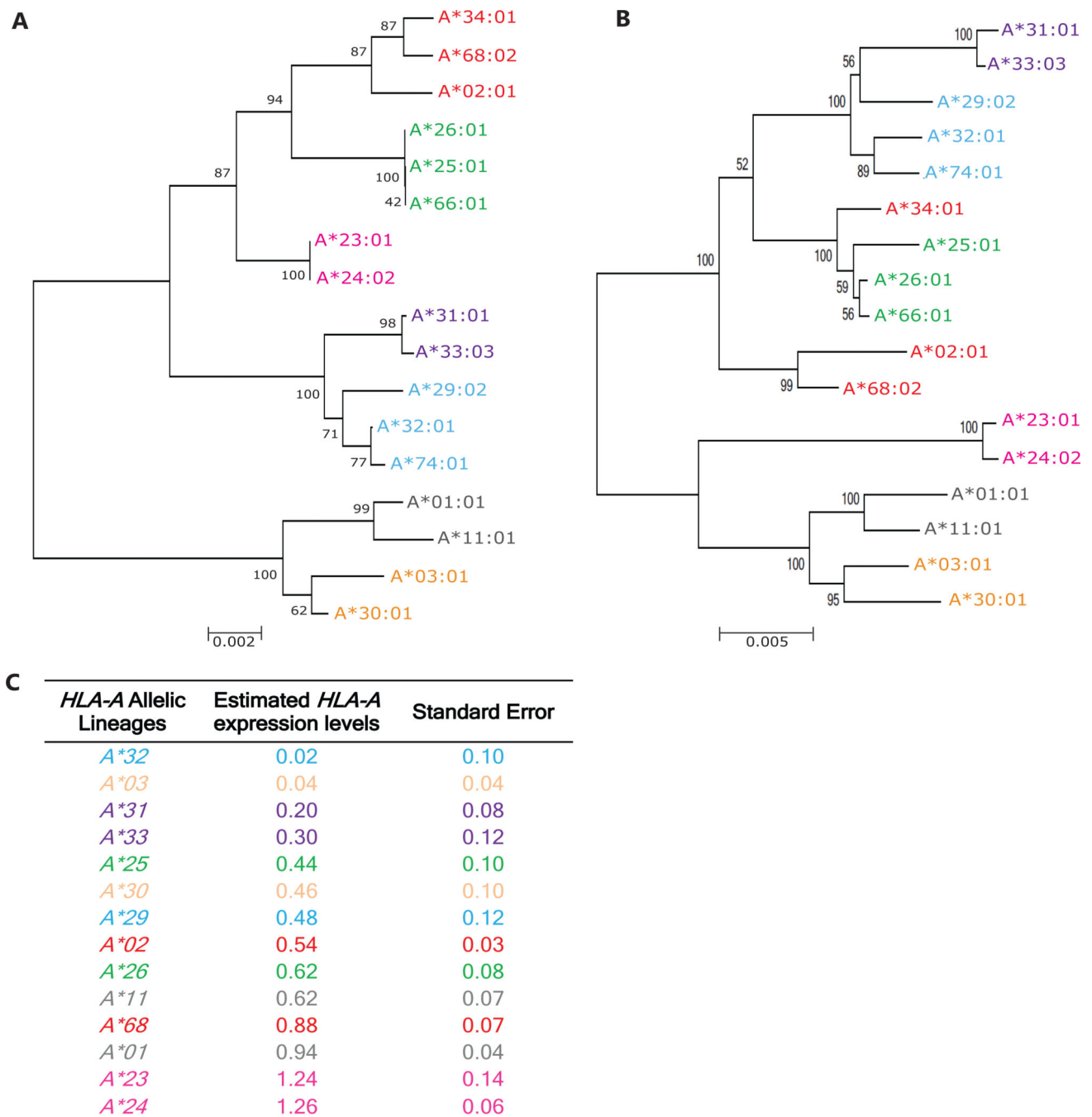
sliding window analysis of aligned sequences consisting of a single representative of each allelic lineage (n=27). The x axis represents the nucleotide position in the alignment relative to the TSS (represented as 0).

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**FIGURE 4.**

Phylogenetic promoter and gene region analysis, and estimated mRNA expression levels for the *HLA-A* gene. Neighbor-joining phylogenetic trees derived from ~2kb of promoter region (**A**) and ~3Kb of gene region (**B**) sequences for 17 lineages within the *HLA-A* gene. Numbers on the nodes represent the bootstrap recovery for that node over 500 replications, and the scale bar represents the number of substitutions along a given branch length. Linear regression estimates of *HLA-A* expression levels for each lineage are calculated from a cohort of 215 EA healthy donors (**C**). Estimated mRNA expression levels and the standard

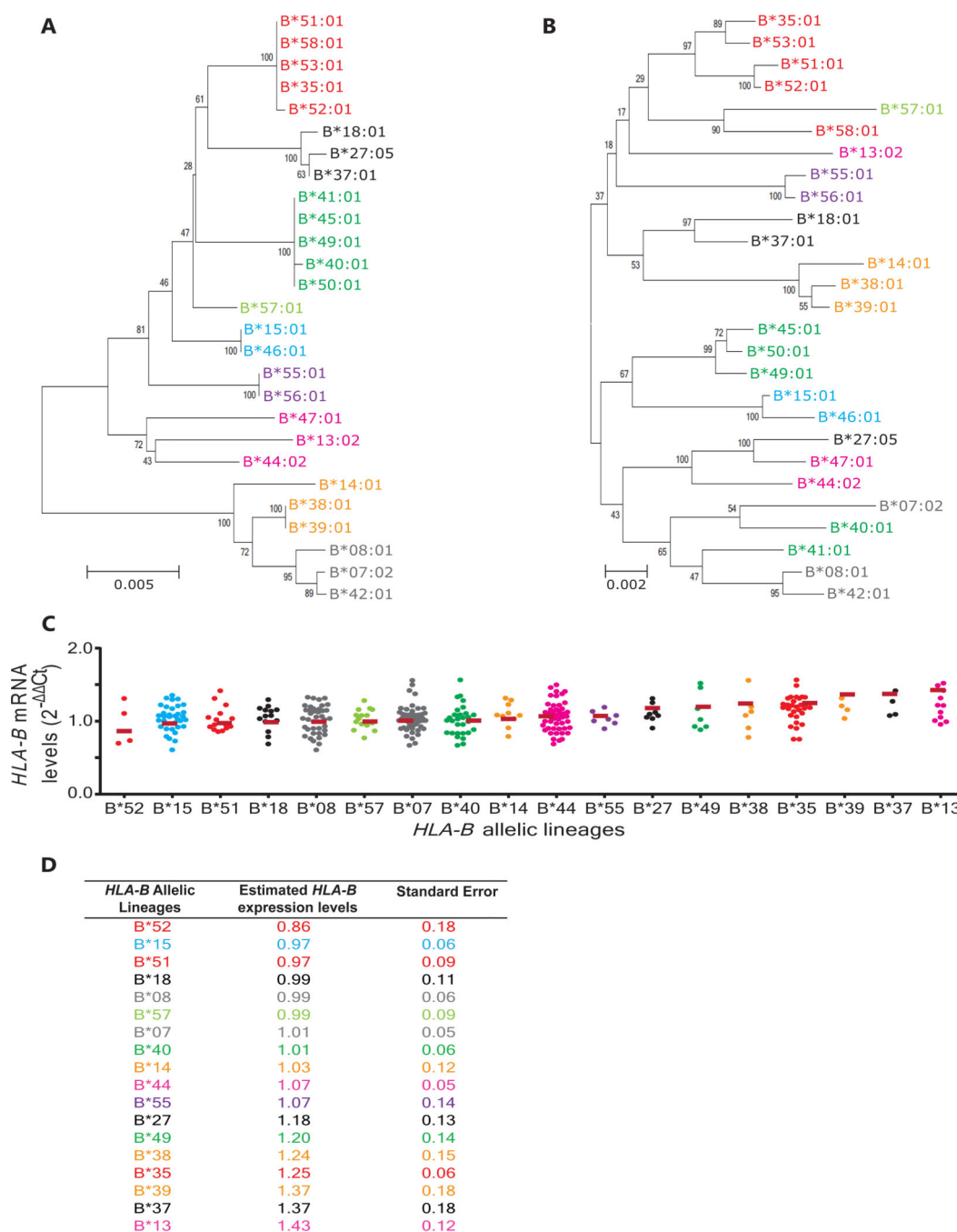
error represent each allelic lineage as a diploid homozygote. Clades within the promoter tree are color coded and maintained in the gene region tree and in the estimated expression levels

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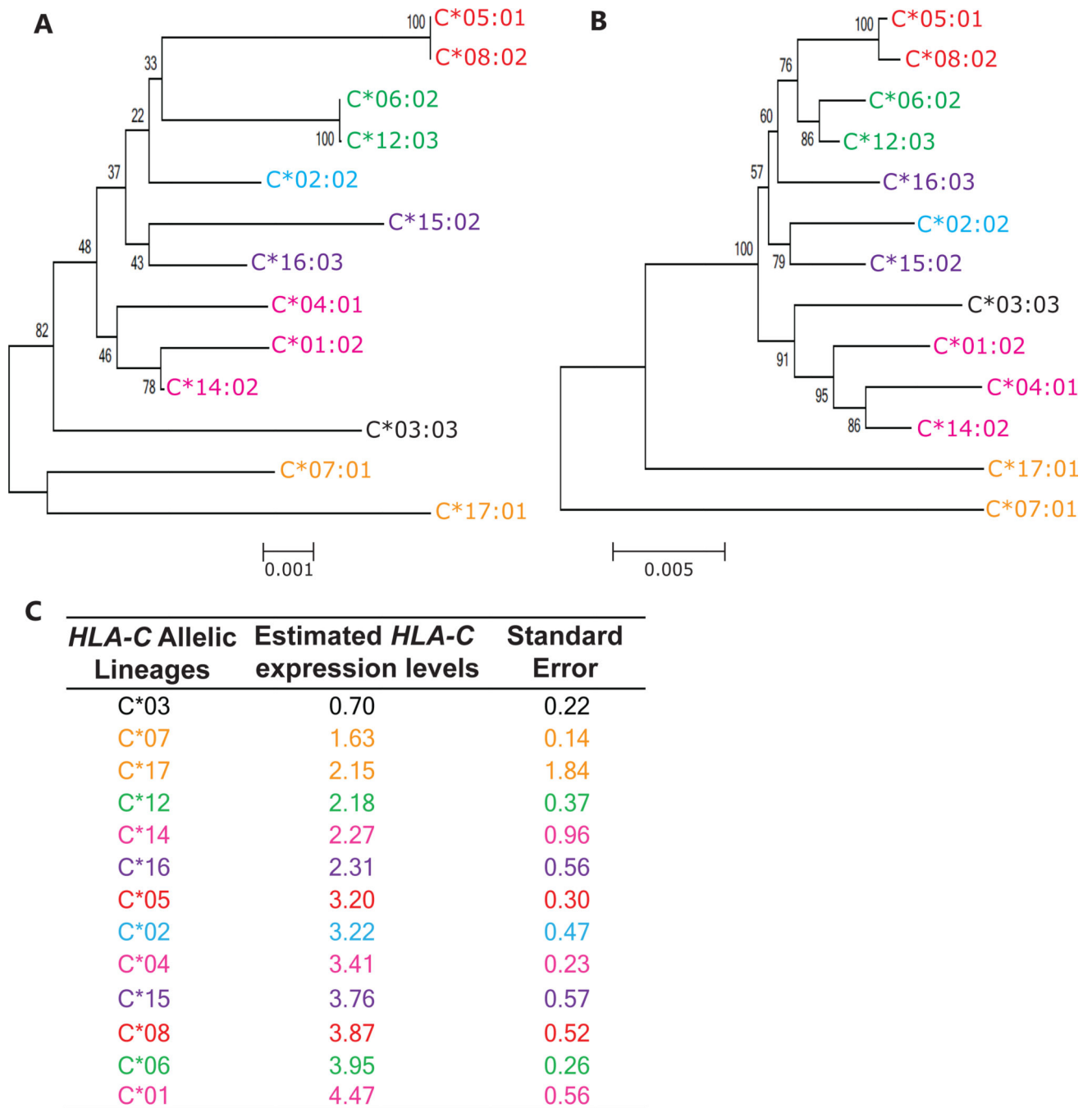
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**FIGURE 5.**

Phylogenetic promoter and gene region analysis, and measured and estimated mRNA expression levels for the *HLA-B* gene. Neighbor-joining phylogenetic trees derived from ~2kb of promoter (**A**) and ~3kb of gene region (**B**) sequences for 27 lineages within *HLA-B* gene. Numbers on the nodes represent the bootstrap recovery for that node over 500 replications, and the scale bar represents the number of substitutions along a given branch length. *HLA-B* mRNA expression levels from 178 EA healthy donors is shown as measured using qPCR (**C**). *HLA-B* mRNA levels were calculated using the $2^{-\Delta\Delta C_t}$ method and *B2M*

as the housekeeping control. Relative expression levels are plotted twice for each donor, i.e. once for each lineage present. Lineages found in ≥ 4 individuals are shown, and the horizontal line represents the estimated *HLA-B* expression levels for each lineage as calculated using linear regression. Estimated mRNA expression levels and the standard error represent each allelic lineage as a diploid homozygote (**D**). Clades within the promoter tree are color coded and maintained within the gene region tree, and in the measured and estimated expression levels

**FIGURE 6.**

Phylogenetic promoter and gene region analysis, and estimated mRNA expression levels for the *HLA-C* gene. Neighbor-joining phylogenetic trees derived from ~2kb of promoter (**A**) and gene region (**B**) sequences for 13 lineages of the *HLA-C* gene. Numbers on the nodes represent the bootstrap recovery for that node using 500 replications, and the scale bar represents the number of substitutions along a given branch length. Linear regression estimates of *HLA-C* mRNA expression levels for each lineage were calculated from a cohort of 256 EA healthy donors (**C**). Estimated mRNA expression levels and the standard error

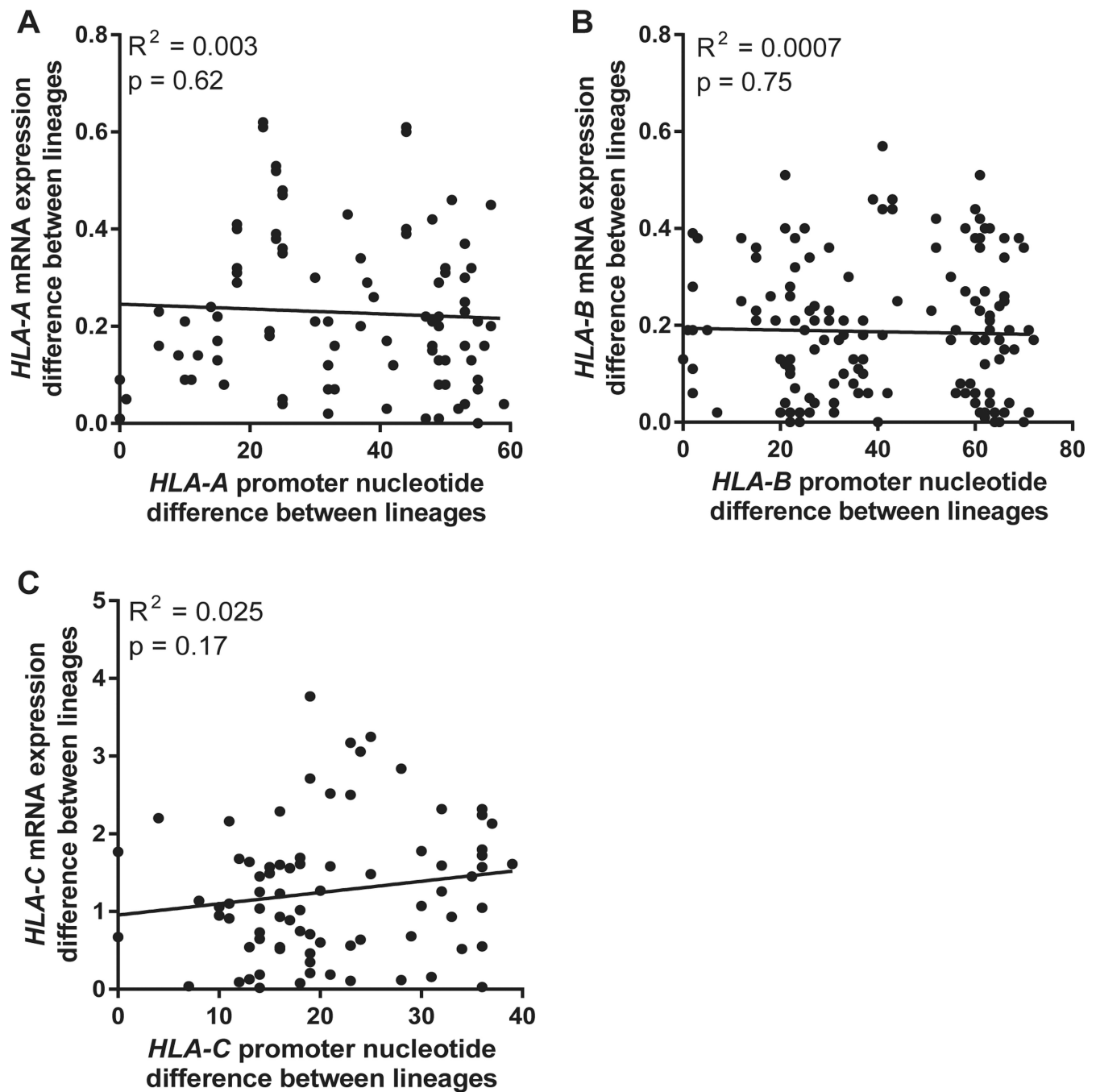
represent each allelic lineage as a diploid homozygote. Clades within the promoter tree are color coded and maintained in the gene region tree and in the estimated expression levels

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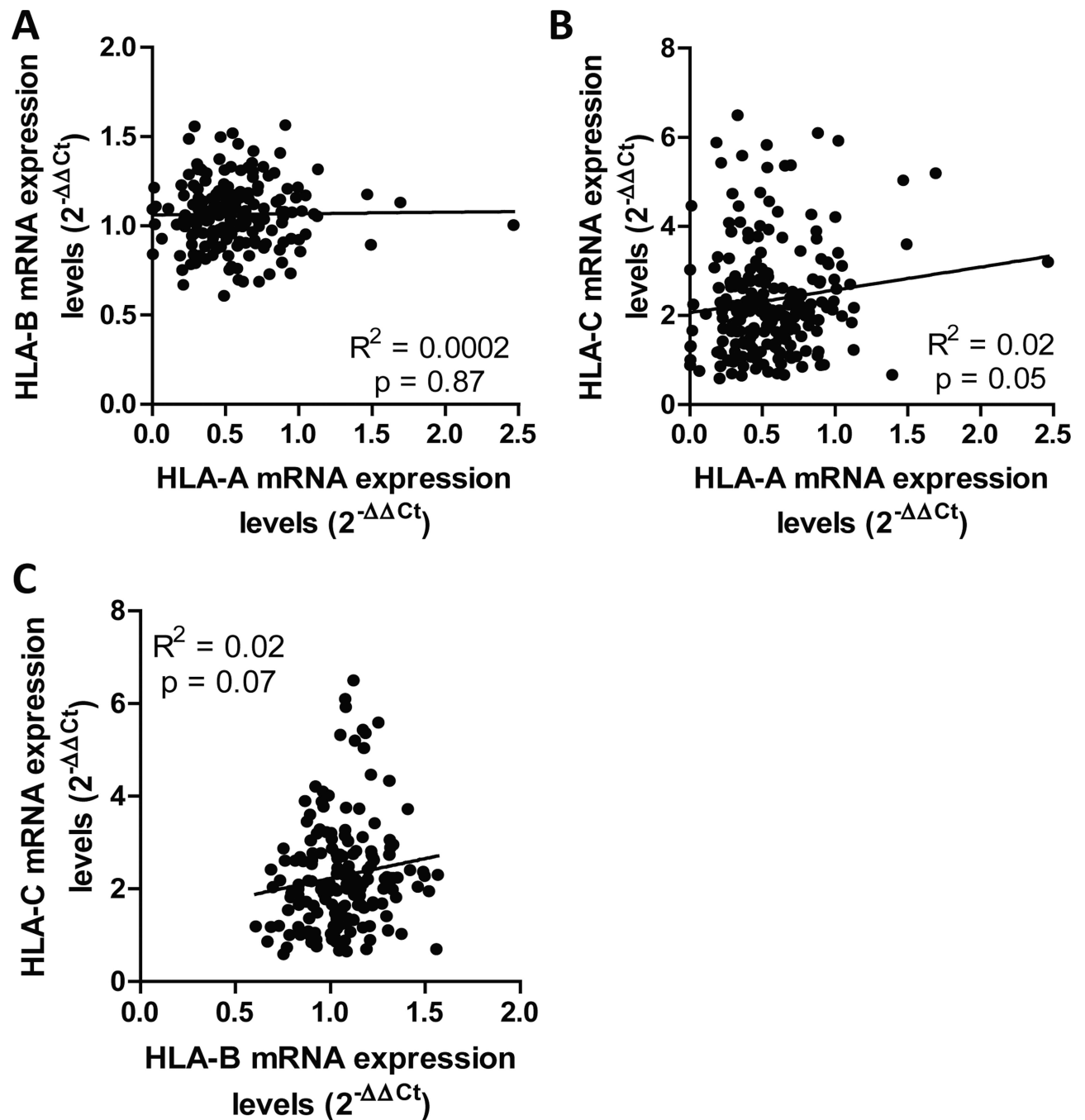
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**FIGURE 7.**

The number of nucleotide differences in the promoter region does not correlate with change in mRNA expression level between pairs of lineages at the classical class I loci. The number of promoter region nucleotide differences was tested for a correlation with the difference in expression levels across all pairs of lineages at the (A) *HLA-A* ($R^2 = 0.003$, $p = 0.62$), (B) *HLA-B* ($R^2 = 0.0007$, $p = 0.75$) and (C) *HLA-C* ($R^2 = 0.025$, $p = 0.17$) loci

**FIGURE 8.**

Lack of a correlation in mRNA expression levels between pairs of classical class I loci. *HLA-A*, *-B* and *-C* mRNA expression levels measured in 172 EA donors show no correlation when comparing (A) *HLA-A* vs. *HLA-B* (Spearman correlation coefficient; ($R^2 = 0.0002$, $p = 0.87$), (B) *HLA-A* vs. *HLA-C* (Spearman correlation coefficient; $R = 0.02$, $p = 0.05$) or (C) *HLA-B* vs. *HLA-C* (Spearman correlation coefficient; $R = 0.02$, $p = 0.07$))